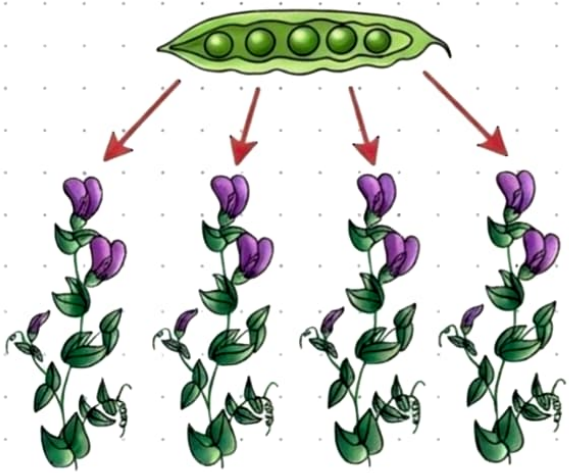
Genetics & Inheritance: Master Study Notes

Exceprt:
"Inheritance is the process by which genetic information is passed from parent to offspring."
- Biology Textbook

Exceprt:
"Gregor Mendel's experiments with pea plants laid the foundations of modern genetics."
- Biology Textbook

Mendel's Laws, Chromosomal Theory, and Genetic Disorders



F_1 generation:
Dominant trait gets expressed.
Always resembles to one of its parents.
Its self-cross leads to few plants ($1/4^{th}$ of total) expressing recessive trait.

F_2 generation:
- $3/4^{th}$ express Dominant trait & $1/4^{th}$ express Recessive trait.
- Do not show any blending.

Conclusions:-

- Mendel called something which was getting passed down through every generation as factors.
- These factors are called as genes.
- Genes are unit of inheritance & contain information required to express a particular trait in an organism.
- Genes which code for a pair of contrasting genes are called as alleles.

E.g: for a characteristic trait i.e. tall, the allelic pair will be TT & tt or Tt.

TT & tt are genotype while tall & dwarf are phenotype.

In a pair of alleles, one is called dominant factor & other is recessive factor.

When the tall & dwarf plants produce gametes, the alleles of parental pair segregate from each other & only one allele is transmitted to a gamete.

identical or homozygous, TT and tt, respectively; the genotype; the phenotype; like the TT parent; factors, one dominates the other; dominant factor while the other factor is recessive; tallness) is dominant over *t* (for dwarfness).

Test cross:-

In a test cross, an organism showing a dominant phenotype (Tt) is crossed with a recessive parent (tt), to check the genotype of the dominant organism.

$$\boxed{\text{i.e. Tt x tt}}$$

Mendel's laws of inheritance:-

Law of dominance:-

- Characters are controlled by discrete units called factors.
- Factors occur in pair.
- In a dissimilar pair of factors one member of the pair dominates the other.

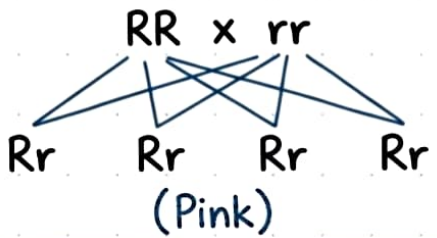
Law of segregation:-

- During zygote formation, the alleles of parents segregate such that the offspring gets a pair of alleles containing 1 factor of each parent.

Incomplete dominance:-

Example: Dog flower / Snapdragon (Antirrhinum).

A cross was made between true breeding red (RR) & true breeding white (rr):-



Resulted in all pink flowers (intermediate of red & white flowers).

A cross was made between genes that can produce a certain enzyme for a substrate S. Following can be the possibilities of the newly formed gene:-

(1) Capable of producing functional enzyme
(2) Capable of producing non-functional enzymes
(3) Capable of producing no enzymes

Codominance:-

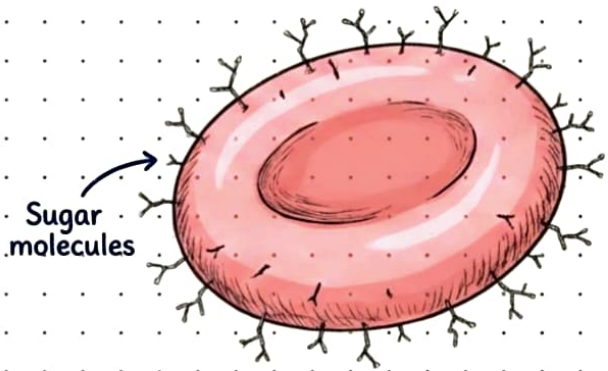
Example: Blood grouping

ABO blood groups are controlled by gene I. Plasma membranes of RBCs has sugar molecules that protrude from its surface and the kind of sugar is controlled by its gene.

The gene (I) has 3 alleles: I^A, I^B, i.

i is a recessive trait & I^A & I^B are both dominant traits.

Thus, I^A & I^B together give sugars of their own types, resulting in AB group of the individual.



Since, more than 2 alleles are controlling a particular gene, thus it is also called as multiple allelism.

Example: Starch producing seed

(BB)

produce starch in large amount

(bb)

does not produce much starch

(Bb)

produce neutral amount of starch

Law of independent assortment:-

When 2 pairs of traits are combined in a hybrid, segregation of one pair of characters is independent of the other pair of characters.

Chromosomal theory of inheritance:-

Mendel's work was not recognised till 1900s:-

1. Communication was not easy & his work was not widely published.
2. His concept of factors as stable & discrete units that controlled the expression of traits & of the pair of alleles which did not blend with each other was not accepted by his contemporaries as an explanation.
3. Mendel's approach of using maths to explain a biological phenomenon was totally new & unacceptable to many biologists.
4. Mendel's work suggested that factors were discrete units, he could not provide any physical proof for its existence.

Boveri & Sutton's Chromosomal Theory

By 1902, Boveri & Sutton studied the movement of chromosomes during meiosis in cell division.

- They also noted that behaviour of chromosomes was similar to behaviour of genes.
- Sutton & Boveri argued that the pairing & separation of a chromosome would lead to the segregation of a pair of factors.

Morgan's Work with Drosophila

Thomas Hunt Morgan worked with fruit flies called *Drosophila melanogaster* :-

(i) They can be easily grown in lab.

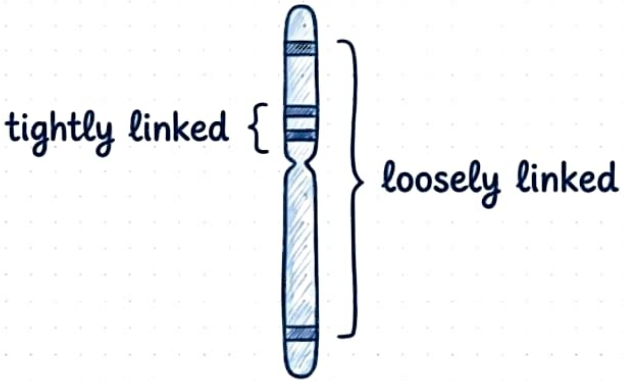
(ii) They complete their life cycle in about 2 weeks.

(iii) They can produce large number of flies in a single mating.

(iv) Male & female sexes are easily distinguishable.

(v) They have various hereditary variations.

Linkage & recombination:-



When 2 genes are located on X chromosome, some are very tightly linked & have very less distance among them & thus they don't show recombinants but mostly parental genes.

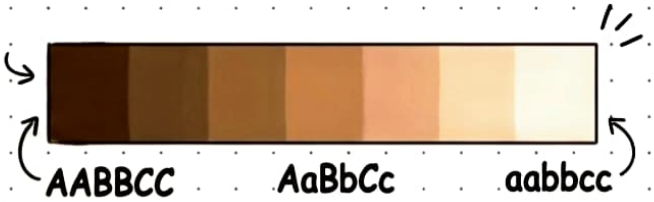
However, some are loosely linked & have more distance away then & thus show recombinants.

Alfred Sturtevant used frequency of recombinants as a measure of distance between those genes & mapped their position on chromosome.

Polygenic inheritance:-

Traits which are controlled by 3 or more alleles are called as polygenic traits.
E.g: Human skin colour.

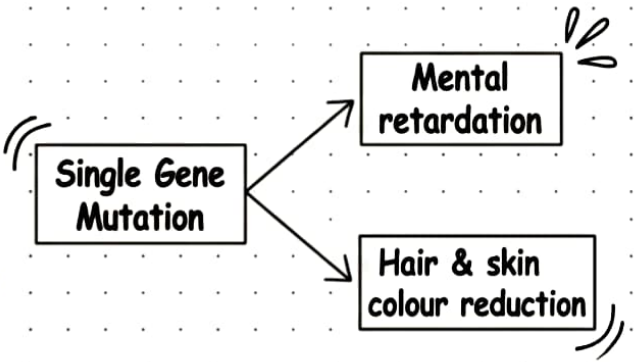
Suppose 3 genes A, B & C control this trait with dominant alleles AABBCC produce the darkest complexion possible & aabbcc produce the lightest complexion possible. Thus the genotype with 3 dominant & 3 recessive traits will have intermediate skin colour.



Pleiotropy:-

A single gene can control multiple traits. It affects metabolic pathways to produce distinct phenotypes.

E.g: phenylketonuria, it is caused by mutation in gene that codes for enzyme phenylalanine hydroxylase. This leads to mental retardation & reduction in hair & skin colour.



Sex determination:-

Henking traced a special type of nuclear structure present in 50% of sperms after spermatogenesis. He gave it a name X-body. Later, it came to be known as X-chromosome.

Insects:-

- XO type chromosomes present.
- Some sperms bear X-chromosome & some not.
- Eggs fertilized by sperms bearing X-chromosome becomes females & other becomes males.
- Y chromosome was called as sex chromosome & other as autosomes.
- E.g: Grasshopper is an example of XO type in which males have 1 sex-chromosome & autosome but females have both X-chromosome.

Male Heterogamety	Female Heterogamety
This autosome chromosome came to be known as Y-chromosome. Same occurs in Drosophila also. This is called as male heterogamety.	Female heterogamety can be found in birds where females have ZW pair but males have ZZ chromosomes.

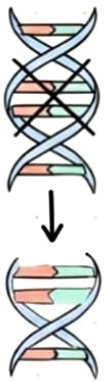
In honeybees, fertilized egg develops as female with 32 chromosomes whereas unfertilized eggs develop as males (drones) with 16 chromosomes, which later develop by parthenogenesis (mitosis).

Among the males an X-chromosome is present but its counter part is distinctly smaller and called the Y-chromosome; Drosophila the males have one X and one Y chromosome, whereas females have a pair of X-chromosomes besides autosomes; male heterogamety; But two different types of gametes in terms of the sex chromosomes.

Mutation:-

It is a phenomenon which results in alteration of DNA sequences & consequently results in changes in the genotype & phenotype of the organism. It also leads to variation in DNA. Loss / gain of a segment in DNA results in mutations.

Substitution: Incorrect base gets matched. Point mutation.



Insertion/frame shift mutation: Extra base inserted inserted between genes.

Deletion: Deletion of base from genes.

Mutation is a phenomenon which results in alteration of DNA sequences and consequently results in changes in the genotype and the phenotype of an organism; loss (deletions) or gain (insertion/ duplication) of a segment of DNA, result in alteration in chromosomes; located on chromosomes; alternative.

Substitution; is called the point mutation; of a particular.

Pedigree Analysis Symbols

Male

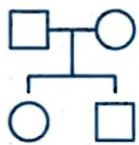
Female

Sex unspecified

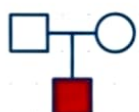
Affected individuals

Mating

Mating between relatives (consanguineous mating)



Parents above and children below



Parents with male child affected with disease

Five unaffected offspring

Colour blindness:-

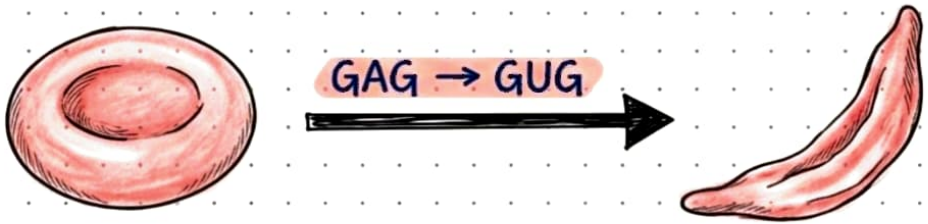
- Sex-linked recessive disorder.
- Defect in either red/green cone of eye due to which the individual does not able to differentiate between red & green colour.
- Occurs due to mutation in X chromosomes.
- Occurs in 8% of males & 0.4% females.

Haemophilia:-

- Sex linked recessive disorder.
- Blood clotting protein molecules get affected.
- A normal cut on the person's body leads to non-stop bleeding.
- Queen Victoria was a carrier of this disease.

not herself colour blind because **the gene is recessive; will not normally be colour blind**, unless her **mother is colour blind** and her **father is colour blind**;
Haemophilia : This **sex linked recessive disease**; the cascade of proteins involved in the clotting of blood; in an affected individual a **simple cut** will result in **non-stop bleeding**; The heterozygous female (carrier) for haemophilia may transmit the disease to sons; rare because mother of such a female has to be at least carrier and the father should be haemophilic; pedigree of Queen Victoria shows a number of haemophilic descendents as she was a carrier of the disease.

Sickle-cell anaemia:-



- Autosome linked recessive disease.
- Controlled by a pair of alleles Hb^A & Hb^S
- Individuals with Hb^S Hb^S show this disease & Hb^A Hb^S are carriers.
- It causes due to substitution at 6th position of beta-globin chain i.e glutamic acid → valine i.e GAG → GUG. This results in changing the shape of RBC molecules from biconcave disks to sickle like structures.

Sickle-cell anaemia : This is an autosome linked recessive trait; can be transmitted from parents to the offspring when both the partners are carrier for the gene (or heterozygous); single pair of allele, Hb^A and Hb^S. Out of the three possible genotypes only homozygous individuals for Hb^S (Hb^S Hb^S) show the diseased phenotype; Heterozygous (Hb^A Hb^S) individuals appear apparently unaffected but they are carrier of the disease as there is 50 per cent probability of transmission of the mutant gene to the progeny, thus exhibiting sickle-cell trait; The defect is caused by the substitution of Glutamic acid.

Phenylketonuria:-

- It is an autosomal recessive trait.
- The individual lacks an enzyme that converts amino acid phenylalanine → Tyrosine. Thus, phenylalanine → phenylpyruvic acid & gets accumulated leading to...

Thalassemia:-

- Autosome linked recessive disease.
- It occurs due to either insertion / deletion in one of the globin chains.

alpha-Thalassemia:

- production of alpha-globin chain is affected.
- controlled by HBA1 & HBA2 at 16th position.

beta-Thalassemia:

- production of beta-globin chain is affected.
- controlled by HBB at 11th chromosome position.

- The more affected genes the less globin chains will produce. Thus leading to anaemia.

Chromosomal disorders:-

Failure of segregation of chromatids during cell division results in gain / loss of a chromosome called aneuploidy.

Failure of cytokinesis after telophase stage of cell division results in increased number of chromosomes called polyploidy.

Down's syndrome:-

- Presence of an additional copy of chromosomes at 21st position (trisomy 21) leads to this disease.
- Individual is short statured, round & furrowed head & tongue and partially open mouth.

Klinefelter's syndrome:-

- Additional copy of X chromosome is present at 47th position.
- Individual has masculine development but feminine characteristics also present. They are sterile.

Turner's syndrome:-

- Absence of X chromosome at 45th position.
- They have lack of secondary sexual characters and these females are sterile.

Genetic Disorders

Mendelian Disorders

- Colour Blindness (Sex-linked recessive)
- Haemophilia (Sex-linked recessive)
- Sickle-cell Anaemia (Autosome linked recessive)
- Phenylketonuria (Autosomal recessive)
- Thalassemia (Autosome linked recessive)

Chromosomal Disorders

- Aneuploidy / Polyploidy (Cell division failure)
- Down's Syndrome (Trisomy 21)
- Klinefelter's Syndrome (47th position, XXY)
- Turner's Syndrome (45th position, XO)